

DoomSat: playing Doom through a real mission stack, with a System One and a System Two

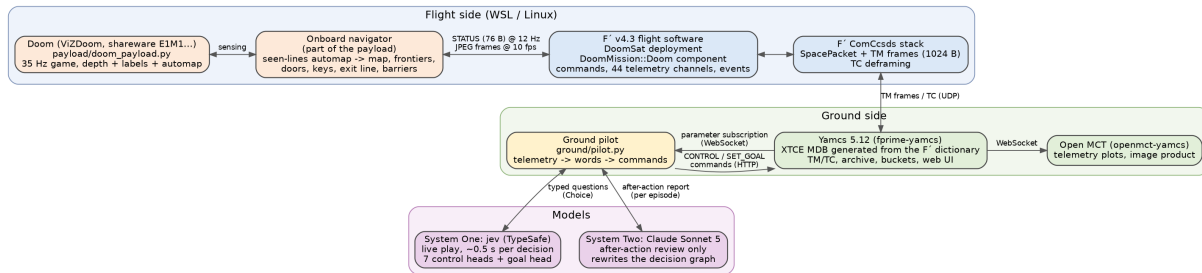
Kevin Horton, 22 September 2026

Built with Claude Code, TypeSafe's *jev*, NASA F', Yamcs, NASA Open MCT and ViZDoom.

Repository: <https://github.com/Devonance/claude-jev-fprime-yamcs-openmct-doom>

1. The idea in one paragraph

Doom is a stress test dressed up as a toy. It is interactive (uplink latency shows up as a character that turns late), high rate (a 320x240 frame every 100 ms has to be fragmented, framed, sent, deframed and reassembled), and self-validating (a dropped frame or a mis-decoded telemetry value is obvious to anyone watching). So we put the game behind a real flight computer and played it from the ground through a real mission control system: the game is a **payload** behind an **F' v4.3** deployment, telemetry and image products go down as **CCSDS** frames into **Yamcs 5.12**, whose mission database is the **XTCE** generated from the F' dictionary, and the ground side plays by uplinking F' commands. The player is not a person: TypeSafe's **jev** answers the moment-to-moment questions (a System One, half a second per decision), and **Claude Sonnet 5** reads an after-action report when an episode ends and rewrites the questions *jev* plays with (a System Two). Code owns the loop. Nobody reads the level file; the character has to find the exit the way a person would.



2. Why two models and not one

A large language model in a 35 Hz control loop is the wrong tool: 5 to 10 seconds per reply, and a reply that reasons about the whole situation when the situation only asks "turn left or right?". *jev* is the opposite: it answers a handful of typed multiple-choice questions in one forward pass in about 400 ms, and it gives a confidence per answer. That is the shape of a reflex. What it cannot do is notice, over many decisions, that a question is badly worded or a threshold is wrong. That

is the shape of a review, and it is what the slow model is for. The split is therefore by *timescale*, not by importance: jev plays every tick; Sonnet is consulted only between episodes, with a two-kilobyte report and the current graph, and returns a revised graph that code validates before the next episode uses it. Nothing slower than jev sits in the live loop.

This is the same pattern as the earlier Jezero rover demo (jev drives, Claude plans), moved onto a flight and ground stack and given a harder honesty rule.

3. What the system actually does

3.1 The world

The game is shareware Doom, level E1M1 onward, run by ViZDoom 1.3 at 35 Hz inside WSL. Finishing a level does not end the run: the payload advances to the next map and carries the weapons and ammunition over, as the game does (keys stay behind). Dying restarts the same level; the map the navigator built is kept, as a person's memory would be. Freedoom is the fallback when the shareware IWAD is not wanted.

3.2 What the player side may see

The rule: jev and code go in blind, like a person who knows how to play Doom but has never seen this level. The level file is never read by the payload or the models. What replaces the person's eyes:

Sense		What it stands in for	Source
Depth	buffer (range camera)	where the walls are	ViZDoom depth buffer, calibrated: 7.16 map units per step, perpendicular distance
Object labels		recognising monsters, pickups, keys, barrels	ViZDoom labels buffer, only things in view
In-game seen lines only	automap, Tab	the map a player sees when pressing Tab	ZDoom automap in NORMAL mode: only lines the player has looked at, in the engine's default categories (wall, floor step, ceiling change = door, locked door in the colour of its key, exit line)
HUD variables		health, armor, ammo, position, heading	ViZDoom game variables

Only the automap's colours are changed, so that code can read the categories off the pixels; no category is added and no unseen line is drawn. Not used: the automap's whole-map or show-objects modes, the "show trigger lines" option, sector or line geometry from the game state, monster counts, item lists, cheats. A WAD statistics script exists for the developer (choosing which levels to demo, checking results) and is never imported by the payload.

3.3 The onboard senses (there is no planner)

The first version of the payload carried a route planner: a grid, a Dijkstra search over the automap lines, frontier targets, door and switch handling. It was replaced. Every failure it produced was a sensing error dressed up as a plan, and a planner hides those from the model that is supposed to be playing. The payload now only senses and remembers:

- it stamps the automap into a world raster (4 map units per pixel) and sweeps the floor it has seen with the range camera;
- it counts where the player has walked, per 32-unit cell;
- for eight directions around the player (every 45 degrees, three rays each) it reports how far the map is open and whether the ground that way is unexplored (never seen), new (seen, not walked), partly walked, or walked before;
- it reports what is at arm's length ahead: a wall, a door, the exit switch, a locked door, a monster or barrel, or something the map does not show (barred windows and fake doors, which a depth camera sees but the automap never draws);
- it remembers where an exit line, a key and pickups were seen, and whether the player is stuck;
- what it bumps into becomes a barrier for the rest of the level; barrels and pillars become barriers from their labels.

The map of a level is kept across attempts, as a player remembers a layout after dying; the game itself restarts from the beginning with everything in it.

3.4 The flight software and the link

The F' deployment is a fprime-bootstrap project with one custom component, `DoomMission::Doom`, on a 20 Hz rate group. It is a TCP client to the payload: a 76-byte status record twelve times a second becomes 44 telemetry channels, and each JPEG frame is split into 960-byte `FrameChunk` records that go straight into the com queue as telemetry packets (bypassing the channel sampler, which would drop most of them). Five commands go the other way: `CONTROL` (move, strafe, turn in degrees, fire, use, weapon), `SET_GOAL`, `RESET_GAME`, `FRAME_RATE`, `EXPLORE_HINT`. Events mark episodes, levels, keys, goals and dropped frames.

Downlink is the stock F' `ComCcsds` stack: Space Packets (APID 1 telemetry, 2 events, 3 files) in fixed 1024-byte TM frames over UDP. The `fprime-yamcs` launcher generates the XTCE from the F' JSON dictionary (148 parameters, 54 commands), starts Yamcs with it and bridges the frames. Yamcs decodes, archives and serves the parameters over its WebSocket and HTTP APIs, which is where the ground pilot, Open MCT and the dashboard read them.

3.5 The ground pilot and jev's decision graph

`ground/pilot.py` subscribes to the Doom channels, reassembles frames, and every ~0.5 s turns the numbers into words and asks jev. jev never sees a number: a route bearing of 37° is "left"; a clearance of 60 units is "tight"; a nearest enemy at 120 units is "close". The questions are the *heads* of the graph, each a typed choice with criteria, and code decides which options are on the menu (Backward only when the space behind is known clear, strafes only when a side is clear, Use only when something is at arm's length):

jev is a System One model: it classifies a structured state document in one forward pass and returns, per question, an option with probabilities (Choice) or a yes-probability (Noul). It does not reason,

plan or remember. So the graph is written the way such a model is meant to be used: one narrow question per head, options with contrastive criteria (what it covers, what it is not for, examples) that reference fields of the state, yes-no questions as Nouls, and code combining the answers.

Head	Type	Options	Decides
way	Choice	the open directions among ahead, ahead-left, left, behind-left, behind, behind-right, right, ahead-right, each with its own "now": space and ground	the direction; jev is the navigator
advance	Noul	yes / no	walk forward this tick
use	Noul	yes / no	press Use (asked only with a door, the exit or a locked door at arm's length, or when stuck)
fire	Noul	yes / no	only with a living enemy in the crosshair and ammunition
dodge	Choice	Carry on / Dodge left / right / back (only the open sides are offered)	evasion from a close enemy
turn	Choice	Hard left ... Hard right / Turn around	aim at the enemy when the goal is to fight
weapon	Choice	Keep / Pistol / Shotgun	ammunition scarcity
goal (every 8th tick)	Choice	Kill enemies / Restore health / Stock ammo / Add armor / Explore	what to prioritise

The answers map onto one **CONTROL** command: the way becomes a turn setpoint of 0, 45, 90, 135 or 180 degrees (or a step backward when the way behind is open), advance becomes the move, and the option probabilities give hysteresis: the direction only changes when jev is clearly surer of the new one than of the last. A large turn is allowed to finish before the next question is asked, or every half second would re-issue it.

Turning is an onboard setpoint (the command says "turn 90 degrees", the payload executes it at six degrees per tic), because a held turn rate with a half-second decision loop over-rotates.

3.6 System Two: the minute bump and the after-action review

System Two is consulted on two clocks. Every minute the pilot sends Sonnet a progress check: seconds into the level attempt, map cells gained in the last minute, position and heading, what the navigator is doing, the walk of the last two minutes (most visited 64-unit bins) and the navigator's map product rendered as text (walls, seen floor, walked cells, unexplored edges, the player). Sonnet answers with a compass bearing, how long to hold it and a goal; the pilot sends them as **EXPLORE_HINT** and **SET_GOAL**, and the navigator re-picks its frontier with the hint in force. That is the bump out of a random walk.

Each level attempt has a budget of three minutes. If the level is not finished by then, the pilot resets the game (**RESET_GAME**); the payload starts a new episode on the same level, keeping the map

it built, and the episode boundary triggers the after-action review. When an episode ends (death, level finished, budget spent, run end) code writes a report of about two kilobytes: outcome, level, duration, decisions, health trace, damage taken with no enemy in view, cells explored, distance walked, stuck ticks, turn-direction flips, longest hold streak, per-head answer distributions with mean confidence, the navigator modes used, the walk (distinct 128-unit bins, revisit ratio, most visited spots, bumps received) and the last eight decisions in words. Sonnet gets the report and the current graph and returns a revised graph through a JSON-schema call: wording, criteria, thresholds (what counts as blocked, tight, close, critical), the turn sizes, how often the goal is re-asked, the standing order. Code validates the reply (fixed option names, known placeholders, numeric ranges), stores it as `graph_v<N>.json` with the rationale in a changelog, and the next episode is played with it. The Claude CLI is run with non-essential model calls disabled, so the review is one Sonnet call and nothing else.

Three views of the same Yamcs instance: the Yamcs web UI (parameters, command history, events, archive), Open MCT through the `openmct-yamcs` plugin (plots and the image product), and a one-page mission dashboard written for the recording, which shows the reassembled frame, the telemetry, Jev's latest answers with the request latency, Sonnet's latest rationale, F' events, the command archive and the map the navigator built.

4. What each engine is trusted with, and why

Layer	Runs	Trusted with	Not trusted with
Flight code (F' + payload)	35 Hz / 20 Hz	safety (uplink loss $\rightarrow$ hold still), the heading loop, the map, the route, door and switch attempts, level progression	any decision that a person would call judgement
jev (System One)	every ~ 0.5 s	every control decision and the goal, from words	numbers, memory, the level layout
Claude Sonnet 5 (System Two)	every minute, and between episodes	pushing exploration in a direction from the map; rewording and re-tuning the graph from evidence	the controls; it never answers a control head

The reason the map lives onboard and not in jev's prompt is bandwidth and honesty at once: the automap is a picture, the route is a number, and jev should get the one word it needs ("the route goes to the left, space left is clear"), not a map to reason about in 400 ms.

5. Measuring instead of asserting

Things that were measured rather than assumed, and what they changed:

- **The depth buffer is perpendicular distance, 7.16 map units per step.** The first calibration (8.5, radial) put off-centre rays up to 25% short; every wall the camera swept landed inside the room. A rotation test settled it: a wall point keeps its depth times the cosine of its bearing as the view turns.
- **Depth 0 is the sky.** Treating it as "very close" produced phantom walls in every outdoor view.
- **The crosshair is drawn into the depth buffer** (depth 0 at the centre pixel), and the automap crosshair and player arrow are drawn over the map lines beneath them. The arrow rotates with the heading, so the wall pixels under it flickered between stamps and the route flip-flopped between two equal paths every replan. The fix keeps the previously mapped pixels under the arrow.
- **A Yamcs bucket holds 1000 objects.** Frame uploads failed with HTTP 500 once yesterday's products filled it; the products now go into a ring of 20 names.
- **jev's latency** through the whole chain: ~ 450 ms median per decision (eight heads), command issue through Yamcs ~ 45 ms, telemetry at 12 Hz, frames at 10 fps with about one incomplete frame per thousand.
- **A turn commanded and a turn observed are different things.** Subtracting the whole commanded turn from the next bearing, after the telemetry already showed the turn done, made jev turn straight back: hard left, hard right, hard left. Subtracting only the part not yet observed removed the oscillation.

6. What we did today, in order

1. Started from the Freedoom MAP01 setup with a range-camera-only map. It stalled at every door, because a depth camera cannot tell a closed door from a wall.
2. Looked at what other Doom agents do. The System One demo this graph descends from reads the WAD for the exit and the collision grid; the GPT-4/5 work used screenshots plus

a written walkthrough. Neither is blind.

3. Chose the in-game automap in seen-lines mode as the honest map, confirmed in the engine source that its categories are the defaults a person sees, calibrated its scale (`viz_am_scale`) and colours against the palette.
4. Switched to shareware Doom E1M1 onward (the standard demo level), added level progression with weapon carry-over, keys, locked doors, exit lines, switch hunting; added the new telemetry and events to the F' component and rebuilt.
5. Found, one at a time, the sensing mistakes listed in section 5, with code-only self-play harnesses that run the payload in-process at 4x real time.
6. Put jev back in command and watched it fail the way a person would not: walking into the same fake door. Added the steer head so that jev chooses the way out from the four clearances, and fixed the turn compensation.
7. Built the dashboard, the diagrams and this report.

7. Using the pixels: letting jev classify what it sees

The one thing the automap cannot give is what a wall *is*: a plain wall, a door, a switch, an exit sign. A person reads the texture. jev cannot take an image, but it takes numbers and words quickly, so the honest next step is to turn the pixels of a wall segment into a few numbers and ask jev what it is:

- Code already knows, from the depth buffer, which screen columns belong to the surface at arm's length and how far it is. Take that patch of the screen (say 40x60 pixels at the crosshair).
- Reduce it to a short vector a person could have written down: dominant colours (a switch has a small bright panel on a dull background; an exit sign is red with white; a door is a tall uniform slab), edge density (switches and signs are busy, walls are not), vertical symmetry, whether the patch repeats horizontally (a wall texture tiles, a door does not), height of the surface from the depth buffer (doors reach the ceiling, switch panels do not).
- Ask jev one typed question with those values in words: "The surface at arm's length is 64 units wide, reaches the ceiling, mostly grey with a small bright red-and-white rectangle in the middle, busy edges, does not tile. What is it?" with options wall / door / switch / exit sign / monster / pickup, and take the confidence.
- Code acts on the answer the way it now acts on the map categories: door → Use; switch → Use and remember; exit sign → destination; monster → combat heads.

This keeps the division of labour: code measures, jev judges, and the judgement is one 400 ms call per new surface, not one per tick. It would replace both the "walls to try" fallback and most of the reliance on the automap's door colouring, and it is the same trick that made the Jezero rover's science stops work: reduce the picture to a sentence, then ask the fast model.

8. Results

See the run table in the README's status section and `python tools/run_report.py` for the current run. The figures in the table below are from the runs of 22 September 2026 (shareware E1M1, jev live, Sonnet after-action).

Run	jevsions	deci-	Explored cells	Distance walked	Stuck ticks	Levels ished	fin-	Notes
1, no steer head	687		121	27k units	60	0		walked into the fake door repeatedly; turns alternated hard left/right
2, steer head	202		147	7.3k units	0	0		jev chose Left/Right/Back/Turn around in 52 of 202 decisions; turns still alternated
3, steer head + turn compensation fix	see README							

9. Honest limits

- The exit of E1M1 was not reached in the runs recorded here; the character explores the level without getting stuck but has not yet walked into the exit room. The next levers are the ones in section 7 and letting the after-action reviews run for more episodes.
- The automap's door and exit categories are the engine's defaults, but they are still a shortcut compared to reading textures; section 7 is the honest replacement.
- Open MCT loads and connects to Yamcs, but headless Chrome would not paint it for the screenshots; the dashboard is what the recording shows.
- The onboard navigator is more code than the brief wanted. Every piece of it exists because a sensing assumption turned out to be wrong; the list in section 5 is the honest account of that.

10. Sources

- NASA F' v4.3.0: <https://github.com/nasa/fprime>
- fprime-yamcs and fprime-xtce: <https://github.com/fprime-community/fprime-yamcs>, <https://github.com/FarkasJoseph/fprime-xtce/tree/feature/binary-annotation-combined>
- Yamcs 5.12.8: <https://github.com/yamcs/yamcs>
- NASA Open MCT and the openmct-yamcs plugin: <https://github.com/nasa/openmct>, <https://github.com/akhenry/openmct-yamcs>
- TypeSafe jev: <https://docs.typesafe.ai/introduction>, <https://docs.typesafe.ai/sdk/python/api>
- Claude Sonnet 5 through Claude Code: <https://code.claude.com/docs>
- ViZDoom 1.3.0: <https://vizdoom.farama.org>, <https://github.com/Farama-Foundation/ViZDoom>

- Shareware Doom IWAD (Debian doom-wad-shareware), Freedoom: <https://freedom.github.io>
- The System One demo: <https://github.com/sgoedecke/system-one>
- Prior LLM-plays-Doom work: <https://adriandewynter.substack.com/p/will-gpt-4-and-5-run-doom>